Stochastic Calculus and Option Pricing
Lecture 5 Exercise Sheet

5 Week 5 — Self-Financing, Dividend Replication, and Greeks

Throughout, the money-market account is $B_t = e^{rt}$, with constant $r \in \mathbb{R}$. Under the historical measure $\mathbb{P}$, the ex-dividend stock price has dynamics

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{P}}, \qquad \sigma > 0,$$

while one share pays the separate cash dividend $qS_t \,\mathrm{d}t$, with $q \geq 0$. Thus μ is the historical drift of the *price*; the shareholder's gain is $\mathrm{d}S_t + qS_t \,\mathrm{d}t$. Take $q = 0$ when an exercise specifies a non-dividend-paying stock.

Part I — Self-Financing and Dividend-Aware Replication

Exercise 5.1 (Medium — Derive self-financing from rebalancing)**.** First set $q = 0$. Trade at dates $0 = t_0 < \cdots < t_n = T$, holding Δ_i shares and β_i bank-account units on $(t_i, t_{i+1}]$. Let $X_i = \Delta_i S_{t_i} + \beta_i B_{t_i}$. Rebalancing takes place at the *observed* prices at t_{i+1}.

(a) Write the portfolio value just before and just after rebalancing at t_{i+1}. If no cash enters or leaves, show that

$$S_{t_{i+1}}(\Delta_{i+1} - \Delta_i) + B_{t_{i+1}}(\beta_{i+1} - \beta_i) = 0.$$

What pays for an increase in the stock holding?

(b) Deduce the exact wealth increment $X_{i+1} - X_i = \Delta_i(S_{t_{i+1}} - S_{t_i}) + \beta_i(B_{t_{i+1}} - B_{t_i})$. Explain why the old holdings appear.

(c) Pass to continuous time and state the self-financing gain equation

$$\mathrm{d}X_t = \Delta_t \,\mathrm{d}S_t + \beta_t \,\mathrm{d}B_t.$$

If Δ, β are continuous finite-variation processes, also apply the product rule to $X_t = \Delta_t S_t + \beta_t B_t$ and identify the equivalent rebalancing constraint.

(d) Restore the dividend yield q. Write the gain of one share and the self-financing equation. Explain why changing Δ without a compensating cash-account trade would require outside money.

Exercise 5.2 (Medium — Derive the PDE for a dividend-paying stock)**.** Let $V_t = v(t, S_t)$, where $v \in C^{1,2}([0,T) \times (0,\infty))$, and let a self-financing replicating portfolio hold Δ_t shares and β_t bank account units.

(a) Apply Itô's formula to V_t under the historical price dynamics specified above.

(b) Explain why the portfolio gain is

$$\mathrm{d}X_t = \Delta_t\big(\,\mathrm{d}S_t + qS_t\,\mathrm{d}t\big) + \beta_t\,\mathrm{d}B_t.$$

(c) Match the Brownian coefficients to identify Δ_t. Then use $\beta_t B_t = v(t,S_t) - \Delta_t S_t$ to derive

$$\boxed{v_t + (r-q)sv_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0.}$$

Locate precisely where the historical price drift μ cancels.

(d) Redo only the drift matching after incorrectly omitting the dividend gain $q\Delta_t S_t\,\mathrm{d}t$. Which term disappears from the PDE, and what economic cash flow has been lost?

Part II — PDE Applications and No-Dividend Greeks

Exercise 5.3 (Medium — Power payoffs reveal the roles of r and q)**.** For the terminal payoff $g(s) = s^p$, with $p \in \mathbb{R}$, seek a solution of the dividend-adjusted Black–Scholes PDE of the form $v(t, s) = A(t)s^p$.

(a) Derive the ODE for A, including $A(T) = 1$, and solve it.

(b) Check the special cases $p = 0$ and $p = 1$. Interpret the two prices as the value of one unit of cash at T and the value of one share delivered at T, respectively.

(c) Compute the replicating holdings $\Delta_t = v_s(t, S_t)$ and $\beta_t = (v - \Delta S)/B_t$.

(d) For $p > 1$, explain from the formula why curvature introduces a volatility contribution even though the physical drift μ is absent.

Exercise 5.4 (Medium — Derive the no-dividend call Greeks)**.** For this exercise set $q = 0$, $\tau = T - t > 0$, and use

$$C(t, s) = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_1 = \frac{\log(s/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Write $\phi = \Phi'$, and hold the other model parameters fixed in each partial derivative.

(a) Show that $s\phi(d_1) = Ke^{-r\tau}\phi(d_2)$. Compute $\partial_s d_1$ and $\partial_s d_2$.

(b) Differentiate C with respect to s once and twice. Show every density term that cancels, and obtain delta and gamma.

(c) Compute $\partial_\sigma d_1$ and $\partial_\sigma d_2$, then derive vega. Derive rho by differentiating with respect to r.

(d) Differentiate first in τ, then use $\Theta = \partial_t C = -\partial_\tau C$ to derive calendar-time theta. State all five Greeks explicitly.